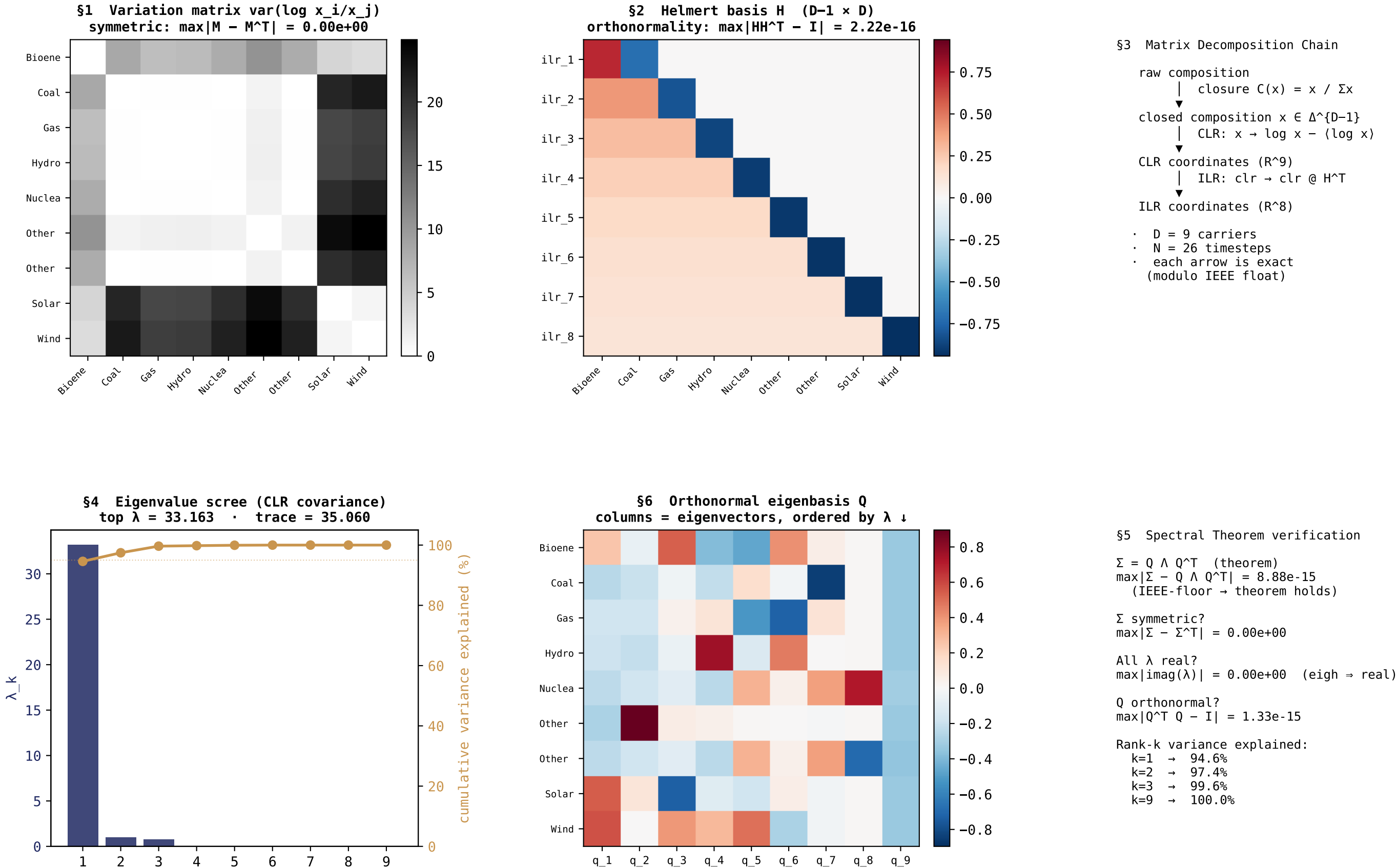


Foundations Plate (Stage 0) – EMB AUS Australia generation TWh.csv · D = 9 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible



Foundations – Numeric Verification – EMB AUS Australia generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

| FOUNDATION    | STATEMENT                                                                        | VALUE     | JUDGMENT              |
|---------------|----------------------------------------------------------------------------------|-----------|-----------------------|
| \$1 Symmetric | $\max  M - M^T $ for variation matrix                                            | 0.000e+00 | exact (= 0)           |
| \$1 Symmetric | $\max  \Sigma - \Sigma^T $ for CLR covariance                                    | 0.000e+00 | IEEE-floor (2.22e-16) |
| \$2 Transpose | $\max  H H^T - I_{\{D-1\}} $ (Helmert orthonormal rows)                          | 2.220e-16 | IEEE-floor            |
| \$2 Transpose | $\max  H^T H - P_{\{D-1\}} $ (projection onto image)                             | 2.220e-16 | IEEE-floor            |
| \$3 Decomp    | closure $\rightarrow$ CLR $\rightarrow$ ILR $\rightarrow$ CLR roundtrip residual | 3.333e-09 | IEEE-floor            |
| \$4 Eigen     | number of eigenvalues found                                                      | 9         | = D (9)               |
| \$4 Eigen     | $\max  \text{imag}(\lambda_k) $ (real because $\Sigma$ symmetric)                | 0.000e+00 | exact                 |
| \$4 Eigen     | $\lambda_1$ (top eigenvalue, variance along leading axis)                        | 33.1632   | –                     |
| \$4 Eigen     | $\text{trace}(\Sigma) = \sum \lambda_k$ (total variance)                         | 35.0596   | –                     |
| \$5 Spectral  | $\max  \Sigma - Q \Lambda Q^T $ (Spectral Theorem residual)                      | 8.882e-15 | IEEE-floor            |
| \$5 Spectral  | $\max  Q^T Q - I $ (orthonormality of eigenbasis)                                | 1.332e-15 | IEEE-floor            |
| \$6 Decomp    | cumulative variance at k=1 (rank-1 approximation)                                | 94.59%    | –                     |
| \$6 Decomp    | cumulative variance at k=2                                                       | 97.45%    | –                     |
| \$6 Decomp    | cumulative variance at k=3                                                       | 99.65%    | –                     |
| \$7 Visualize | Stage-0 Foundations Plate (page 1 of this PDF)                                   | present   | delivered             |

The foundations carry the bedrock. The instrument reads. The expert decides.  
The hashes carry the receipts. The vocabulary holds the line.